BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

General Metric Space

Properties of Metric, Modified Metric, NLS, IPS

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Conducted by

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Motivation of Metric, Norm and Inner Product

Metric Space

Metric Space

A **metric space** is a pair (X, d) where $X \neq \varnothing$ and $d : X \times X \to \mathbb{R}$ satisfies, for all $x, y, z \in X$:

(M1) $d(x, y) \geq 0$;

(M2) $d(x, y) = 0 \iff x = y$;

(M3) $d(x, y) = d(y, x)$;

(M4) $d(x, y) \leq d(x, z) + d(z, y)$.

Here d is called a **metric** on X.

Inequalities in Metric Spaces

Reverse Triangle Inequality

Let (X, d) be a metric space. Show that for all $x, y, z \in X$,

$$|d(x, z) - d(y, z)| \leq d(x, y).$$

Two-point inequality for distances

Let (X, d) be a metric space. Show that for all $x, y, z, w \in X$,

$$|d(x,y) - d(z,w)| \leq d(x,z) + d(y,w).$$

Discrete Metric Space

Discrete Metric

Let X be a non-empty set. Define $d : X \times X \to \mathbb{R}$ as

$$d(x,y) = \begin{cases} 0, & \text{if } x = y; \\ 1, & \text{if } x \neq y. \end{cases}$$

Show that d is a metric on X.

Modified Metrices

Metric via a Fixed Point

Let (X, d) be a metric space and let $x_0 \in X$. Define a function $\delta : X \times X \to \mathbb{R}$ by

$$\delta(x, y) = \begin{cases} d(x, x_0) + d(x_0, y), & \text{if } x \neq y, \\ 0, & \text{if } x = y. \end{cases}$$

Show whether δ is a metric on X or not.

Bounded Transform of a Metric

Let (X, d) be a metric space. Define $d' : X \times X \to \mathbb{R}$ as

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

Show that d' is a metric on X.

Metric via Truncation

Let (X, d) be a metric space. Define $d' : X \times X \to \mathbb{R}$ as

$$d'(x, y) = \min\{1, d(x, y)\}.$$

Show that d' is a metric on X.

Pullback Metric along an Injection

Pullback metric along an Injection

Let (Y, d) be a metric space and let $\alpha : X \to Y$ be an injection. Define $d' : X \times X \to \mathbb{R}$ as

$$d'(x, y) = d(\alpha(x), \alpha(y)).$$

Show that d' is a metric on X.

Isometric Spaces

Isometric Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces. They are said to be **isometric** if there exists a bijection $\alpha : X \to Y$ such that for all $x_1, x_2 \in X$,

$$d_Y(\alpha(x_1), \alpha(x_2)) = d_X(x_1, x_2).$$

Here α is called an **isometry** from X to Y.

Motivation of Norm

Normed Linear Space

Normed Linear Space

A **normed linear space** is a pair $(X, \|\cdot\|)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\|\cdot\| : X \to \mathbb{R}$ satisfies, for all $x, y \in X$ and $\alpha \in \mathbb{R}$ (or $\mathbb{C}$):

(N1) $\|x\| \geq 0$;

(N2) $\|x\| = 0 \iff x = 0$;

(N3) $\|\alpha x\| = |\alpha|\|x\|$;

(N4) $\|x + y\| \leq \|x\| + \|y\|$.

Here $\|\cdot\|$ is called a **norm** on X.

Metric Induced by a Norm

Let $(X, \|\cdot\|)$ be a normed space. Define $d : X \times X \to \mathbb{R}$ as

$$d(x, y) = \|x - y\|.$$

Show that d is a metric on X.

Motivation of Inner Product

Inner Product Space

Inner Product Space

An **inner product space** is a pair $(X, \langle \cdot, \cdot \rangle)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\langle \cdot, \cdot \rangle : X \times X \to \mathbb{R}$ (or $\mathbb{C}$) satisfies, for all $x, y, z \in X$ and $\alpha \in \mathbb{R}$ (or $\mathbb{C}$):

(IP1) $\langle x, x \rangle \geq 0$;

(IP2) $\langle x, x \rangle = 0 \iff x = 0$;

(IP3) $\langle x, y \rangle = \overline{\langle y, x \rangle}$;

(IP4) $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$.

Here $\langle \cdot, \cdot \rangle$ is called an **inner product** on X.

Norm Induced by an Inner Product

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Define $\| \cdot \| : X \to \mathbb{R}$ as

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Show that $\| \cdot \|$ is a norm on X.

Metric Induced by an Inner Product

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Define $d : X \times X \to \mathbb{R}$ as

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle}.$$

Show that d is a metric on X.

Cauchy-Schwarz Inequality

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Show that for all $x, y \in X$,

$$|\langle x, y \rangle| \leq \sqrt{\langle x, x \rangle}\sqrt{\langle y, y \rangle}.$$

or equivalently,

$$|\langle x, y \rangle| \leq \|x\|\|y\|.$$

Thank You!

We'd love your questions and feedback.

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References

[1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.

[2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.